SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Co3 Assessment Tool 1 – Git & Docker Technical Assignment

Course Code:	CSA10	**Course Title:**	Software Engineering
CO Assessed:	CO3	**Assessment:**	Assessment Tool 1 – Git & Docker Technical Assignment
Weightage:	20%	**Total Marks:**	25
CO3 Statement:	Build full-stack applications with an emphasis on containerization, version control, and collaborative development		
Student Name:	Ragul SG	**Reg. No.:**	192525007
Date:		**Duration:**	60 minutes

Code of Conduct

*I **RAGUL SG - 192525007** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student

1. PROJECT VISION AND STAKEHOLDERS

1.1 THE PROJECT VISION

The vision of this project is to transform traditional, energy-blind manufacturing plants into smart, self-optimizing ecosystems using artificial intelligence and the Internet of Things, simultaneously driving down operational costs and carbon footprints. In the **current state**, factories rely on monthly utility bills to gauge energy use, making it impossible to identify which specific machine or shift caused a consumption spike. Maintenance remains reactive, and energy audits are manual exercises that are outdated almost as soon as they are completed. In the **future state**, the platform acts as the central nervous system of the factory's energy grid, providing continuous second-by-second visibility, anticipating energy surges before they happen, and dynamically guiding operations toward maximum efficiency and minimal environmental impact.

The vision rests on four core pillars that directly shape the epics and user stories in Section 2. **Total Visibility** eradicates blind spots in industrial energy consumption down to the individual asset level. **Predictive Intelligence** shifts the organization from reactive reporting to proactive forecasting using machine learning. **Actionable Optimization** gives operators and managers immediate, data-backed recommendations to reduce waste. **Sustainable Compliance** makes green manufacturing an effortless byproduct of operational efficiency rather than a separate compliance burden.

1.2 KEY STAKEHOLDERS, ROLES, AND MOTIVATIONS

A successful Agile backlog is built on a deep understanding of who uses the system and why. Six stakeholder groups were identified, each with distinct pain points and value propositions that map directly to specific user stories later in this report.

Stakeholder	Role	Key Pain Point	What They Value Most
Plant Manager	Oversees daily operations, quotas, and budget	Rising tariffs and unexplained energy bill spikes	Real-time dashboards tying energy cost to production output
Machine Operator	Operates CNC machines, ovens, and other equipment	Disconnected from the financial cost of running machinery	Simple color-coded energy gauges that do not distract from production

Maintenance Engineer	Ensures uptime and repairs failing equipment	Unexpected failures and rigid maintenance schedules	Energy-signature early warnings enabling predictive maintenance
Sustainability Officer	Ensures ESG targets and regulatory compliance	Months of manual data entry for carbon reports	Automated, audit-ready Scope 1 and Scope 2 emission reports
Executive Sponsor	Approves the AI platform budget; answers to the board	Must prove tangible financial return on IoT/AI investment	High-level summaries of dollars saved and tons of CO_2 reduced
IT/OT Integration Manager	Manages IT network and OT systems (PLCs, SCADA)	Security risks from hundreds of new IoT devices	A secure, scalable telemetry pipeline that does not burden the core network

The Plant Manager and Machine Operator anchor the operational value of the platform, the Maintenance Engineer anchors its reliability value, the Sustainability Officer anchors its compliance value, and the Executive Sponsor and IT/OT Manager anchor its strategic and infrastructural feasibility. This spread of motivations is what makes the backlog genuinely cross-functional.

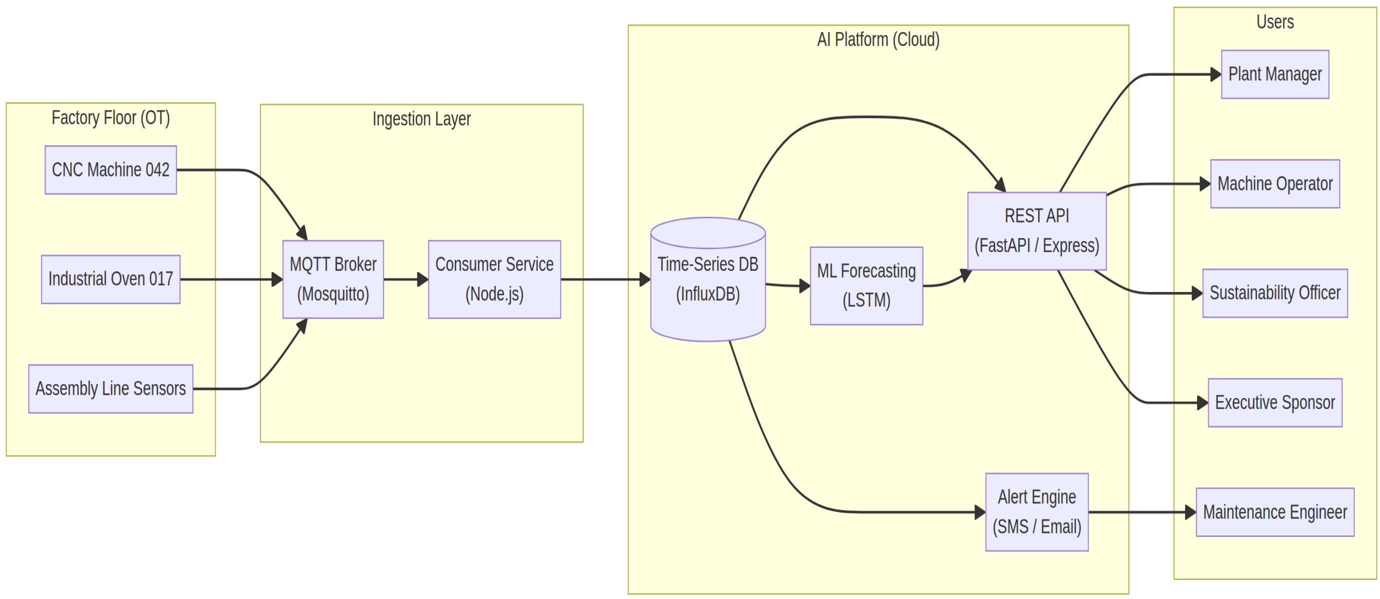


Figure 1 — End-to-end architecture of the platform: factory-floor IoT sensors, the MQTT ingestion layer, the cloud AI platform, and the stakeholder-facing interfaces.

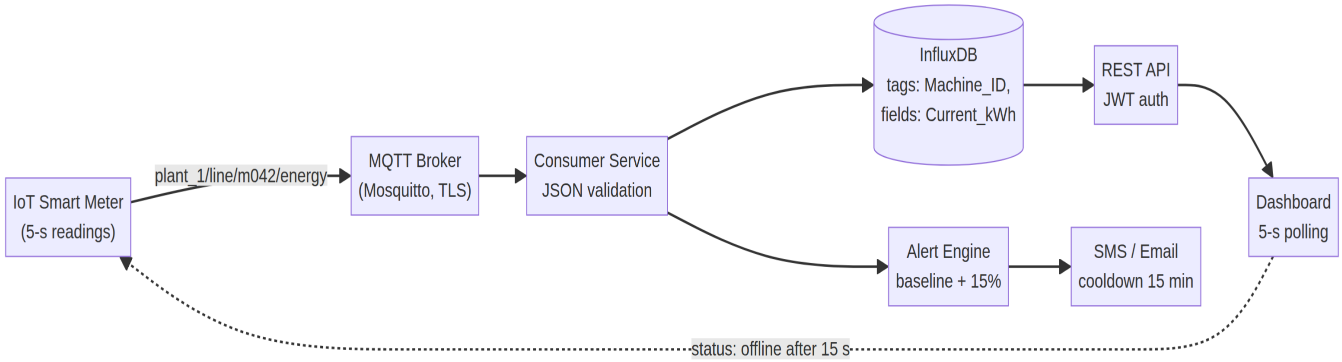


Figure 2 — Data flow for the live monitoring feature: telemetry travels from the 5-second IoT sensor readings through the secured MQTT broker, into the time-series database, and finally to the dashboard, with a parallel branch feeding the alert engine for US 1.2.

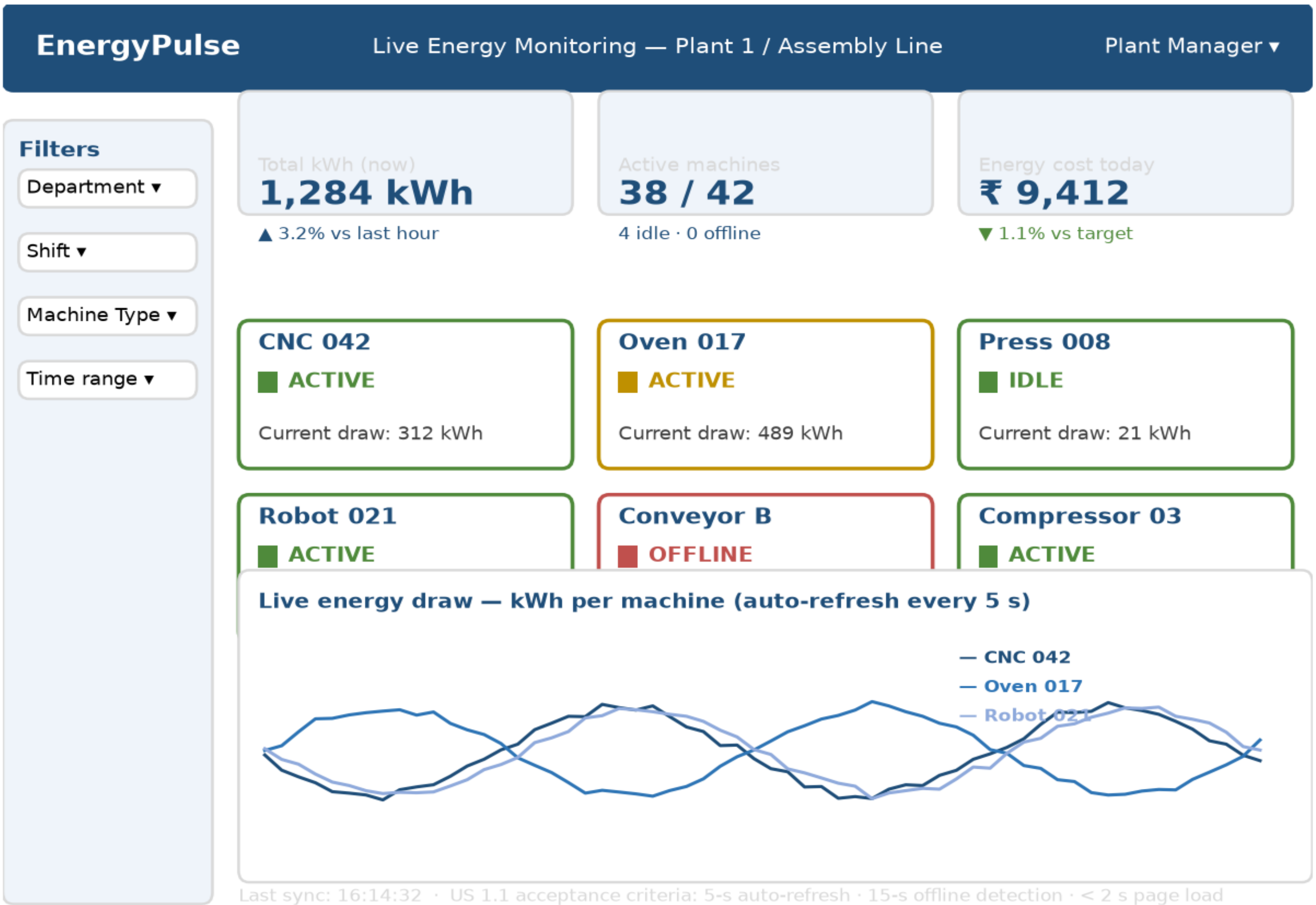


Figure 3 — Wireframe mockup of the live monitoring dashboard: filter sidebar, KPI cards, the color-coded machine status grid, and the auto-refreshing energy-draw chart required by US 1.1 and US 1.2.

2. EPICS AND USER STORIES

The product was decomposed into four epics, progressing from foundational data visibility, through intelligence, toward business-value reporting. Each user story follows the standard *As a... I want... so that...* format and is assigned to the stakeholder whose motivation it serves.

EPIC 1 — REAL-TIME ENERGY MONITORING

This epic establishes the foundational IoT data pipeline and user interfaces required to capture, process, and display live energy consumption metrics from the factory floor. It is the prerequisite for every other epic.

US 1.1 (Core Dashboard): As a Plant Manager, I want to view a centralized real-time dashboard of energy consumption across all machines, so that I can instantly identify areas of excessive energy usage.

US 1.2 (Operator View): As a Machine Operator, I want to view the live energy status of my assigned machine via a simple color-coded gauge (Green/Yellow/Red), so that I can ensure it is operating within its optimal energy limits without distracting me from production.

US 1.3 (Granular Filtering): As a Plant Manager, I want to filter the live energy data by specific departments, shifts, or machine types, so that I can isolate inefficiencies in specific areas of the production floor.

US 1.4 (System Health): As an IT/OT Manager, I want to see connection status indicators for all deployed IoT smart meters, so that I know immediately if a sensor goes offline and stops transmitting data.

EPIC 2 — AI-POWERED PREDICTIVE ANALYTICS

This epic focuses on the platform's core intelligence, utilizing historical data and machine learning algorithms to forecast future energy needs and recommend operational optimizations.

US 2.1 (Demand Forecasting): As a Plant Manager, I want the system to predict peak energy demand for the upcoming week based on inputted production schedules, so that I can plan operations to avoid expensive peak-hour grid tariffs.

US 2.2 (Prescriptive Optimization): As a Plant Manager, I want the AI to recommend alternative machine scheduling combinations, so that I can maintain production quotas while minimizing total energy costs.

US 2.3 (Model Confidence): As a Data Scientist, I want to view the confidence interval and historical accuracy metrics of the prediction models, so that I can continually refine and train the AI for better performance.

EPIC 3 — AUTOMATED ANOMALY DETECTION AND ALERTS

This epic transitions the factory from reactive maintenance to proactive monitoring by using AI to detect deviations from normal energy signatures.

US 3.1 (Instant Notifications): As a Maintenance Engineer, I want to receive automated SMS and email alerts when a machine exhibits sudden, abnormal energy spikes, so that I can inspect it for potential mechanical faults (e.g., motor friction) before it causes unplanned downtime.

US 3.2 (Threshold Configuration): As a Plant Manager, I want to configure the sensitivity (percentage deviation) and time duration required to trigger an anomaly alert, so that I can reduce "alert fatigue" from minor, expected power fluctuations.

US 3.3 (Incident Logging): As a Maintenance Engineer, I want to view a historical log of all triggered energy anomalies alongside their resolution status, so that I can track recurring issues with specific equipment.

EPIC 4 — SUSTAINABILITY AND REPORTING

This epic translates technical energy metrics into business value, providing the documentation required for ESG compliance and executive oversight.

US 4.1 (Compliance Reporting): As a Sustainability Officer, I want to automatically generate monthly carbon emission reports based on consumed energy (Scope 2 emissions), so that I can accurately track our progress toward green manufacturing goals and submit regulatory filings.

US 4.2 (Historical Comparison): As a Sustainability Officer, I want to compare current monthly energy consumption and emissions against the exact same period from the previous year, so that I can measure the long-term effectiveness of our energy-saving initiatives.

US 4.3 (Executive Summary): As an Executive Sponsor, I want to view a high-level summary dashboard displaying total monetary savings and total CO2 tonnage reduced since platform launch, so that I can justify the return on investment (ROI) to the board of directors.

3. ACCEPTANCE CRITERIA

Acceptance criteria were written in BDD-style Given/When/Then form for the three highest-value user stories, covering positive paths, negative scenarios, and non-functional constraints.

US 1.1 — REAL-TIME DASHBOARD

The happy path requires that, given active IoT sensors and a logged-in facility manager, when the user navigates to the Live Monitoring tab, the dashboard displays current kWh usage per machine, updating every five seconds without a manual refresh. The negative scenario requires that when a sensor loses connection, the UI marks that machine as *Offline* (a red icon) after fifteen seconds of missing data. Non-functional constraints require initial page load and rendering within two seconds, and the machine grid must clearly show Machine ID, Current Status (Active/Idle/Offline), and Current Energy Draw in kWh.

US 1.2 — AUTOMATED ALERTS

In the positive scenario, given a machine in active production mode, when its energy draw continuously exceeds the established baseline by fifteen percent for more than two minutes, the system must immediately dispatch a push notification and an SMS alert to the assigned machine operator. The alert-fatigue edge case requires a fifteen-minute cooldown during which no duplicate SMS alerts are sent for the same machine. Content constraints require every alert message body to include the timestamp, machine ID, baseline kW, and recorded spiked kW.

US 2.1 — PREDICTIVE FORECASTS

The positive scenario requires at least thirty days of ingested historical data, after which the Demand Forecast module displays a seven-day interactive predictive graph. With insufficient data, the UI must render a disabled graph with the warning banner *"Insufficient historical data. Minimum 30 days required for forecasting."* Analytical constraints require that hovering over a future data point reveals the predicted peak kW alongside a confidence interval (for example ±5%), with a minimum model accuracy of 85 percent.

User Story	**Scenario**	**Given / When / Then Summary**
US 1.1	Positive (happy path)	Sensors active, user on Live Monitoring tab → kWh per machine updates every 5 s without refresh
US 1.1	Negative (data interruption)	Sensor offline → machine marked Offline (red) after 15 s of missing data
US 1.1	Non-functional	Page renders in < 2 s; grid shows Machine ID, status, and kWh draw
US 1.2	Positive (threshold breach)	Draw exceeds baseline by 15% for > 2 min → push + SMS alert dispatched immediately
US 1.2	Edge case (cooldown)	No duplicate SMS for the same machine for 15 min after an alert
US 2.1	Positive (successful prediction)	≥ 30 days of data → interactive 7-day forecast graph

US 2.1	Negative (insufficient data)	< 30 days → disabled graph with minimum-data warning banner
US 2.1	Non-functional	Tooltip shows predicted peak kW with ±5% confidence; model accuracy ≥ 85%

4. PRIORITIZATION AND PRODUCT BACKLOG

4.1 PRIORITIZATION TECHNIQUE

Prioritization applied the **MoSCoW technique** (Must have, Should have, Could have, Won't have) over the full thirteen-story backlog. Stories classified as *Must have* were admitted into Sprint 1; *Should have* stories were placed in the Sprint 2 candidate pool; and *Could have* and *Won't have* stories were deferred until Sprint 3 or beyond, depending on available velocity and stakeholder feedback collected during the Sprint 1 review. Figure 4 visualizes how the backlog was partitioned across the four MoSCoW categories.

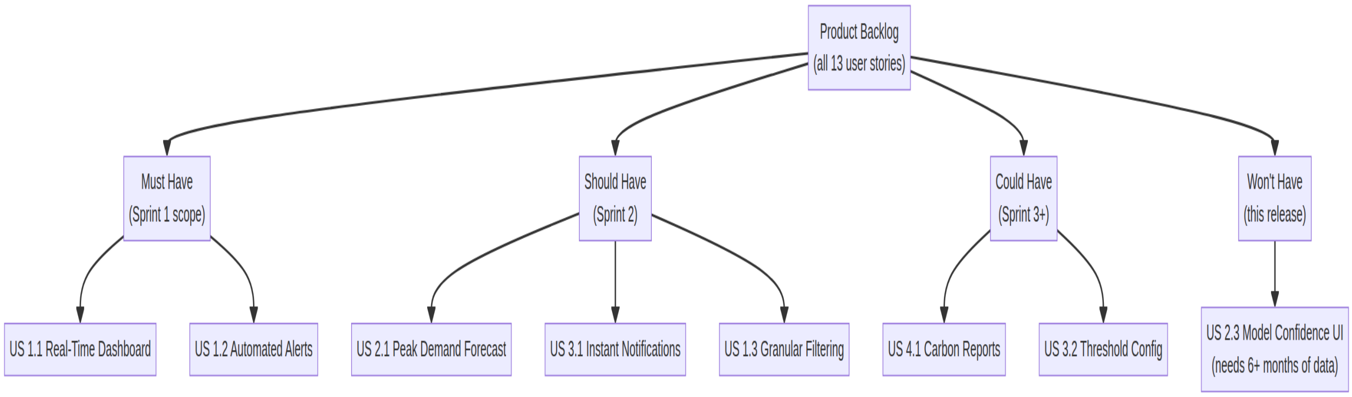


Figure 4 — MoSCoW partitioning of the product backlog: the thirteen user stories distributed across Must/Should/Could/Won't categories, with Sprint 1 scope confined to the Must Have group.

4.2 PRIORITIZED PRODUCT BACKLOG

The backlog was prioritized by business value: foundational data visibility first, then AI optimization, and finally compliance reporting. Dependencies were respected so that no story was scheduled before the data it consumes exists.

Priority	**Epic**	**User Story**	**Business Value**	**Dependency**
1 (High)	1. Monitoring	US 1.1 Real-Time Dashboard	Core foundation; immediate visibility	Hardware/sensor setup

2 (High)	1. Monitoring	US 1.2 Automated Alerts	Prevents critical waste and faults	US 1.1 (data pipeline)
3 (High)	1. Monitoring	US 1.4 System Health	Data trust; early outage detection	US 1.1 (sensor registry)
4 (Medium)	1. Monitoring	US 1.3 Granular Filtering	Targeted root-cause analysis	US 1.1 (data pipeline)
5 (Medium)	2. Prediction	US 2.1 Peak Demand Forecast	Cost savings via load shifting	30+ days of historical data
6 (Medium)	2. Prediction	US 2.2 Utilization Recommendations	Optimizes overall plant efficiency	US 2.1 (AI baseline)
7 (Medium)	3. Anomaly	US 3.1 Instant Notifications	Prevents unplanned downtime	US 1.1 (baseline data)
8 (Medium)	3. Anomaly	US 3.2 Threshold Configuration	Reduces alert fatigue	US 3.1 (alert engine)
9 (Medium)	3. Anomaly	US 3.3 Incident Logging	Audit trail for recurring faults	US 3.1 (alert events)
10 (Low)	4. Reporting	US 4.1 Carbon Emission Reports	Compliance; not operationally critical	US 1.1 (accurate data)
11 (Low)	4. Reporting	US 4.2 Historical Comparison	Year-over-year trend evidence	US 4.1 (12-month data)
12 (Low)	4. Reporting	US 4.3 Executive Summary	ROI proof for the board	US 4.1 and US 4.2

13 (Deferred)	2. Prediction	US 2.3 Model Confidence UI	Model transparency for data scientists	6+ months of training data

The two high-priority stories from Epic 1 (US 1.1 and US 1.2) form the Sprint 1 scope, because they deliver the visible, revenue-protecting core of the product with no upstream dependencies inside the backlog itself. US 1.4 (System Health) was ranked third: it is essential for earning the IT/OT Manager's trust in the pipeline, but it presumes the sensor registry established by US 1.1 and is therefore deferred to Sprint 2.

5. SPRINT BACKLOG AND EFFORT ESTIMATION

5.1 SPRINT BACKLOG TASKS

The sprint backlog decomposes US 1.1 and US 1.2 into five concrete tasks spanning the full stack, from the factory-floor telemetry layer to the browser-facing interface. Each task has a single named owner so that accountability is unambiguous during the daily scrum.

Task	Owner	Scope
Task 1 — Time-Series Database Configuration	Backend	InfluxDB schema separating tags (Machine_ID, Facility_Zone, Machine_Type) from fields (Current_kWh, Voltage, Amperage); retention policies that downsample 5-second data into 1-hour averages after 7 days; Docker-based provisioning with secure credentials
Task 2 — MQTT Ingestion Pipeline	Data Engineering	Mosquitto broker with hierarchical topics (e.g., plant_1/assembly_line/machine_042/energy); a Node.js consumer that validates JSON payloads and writes directly into the database
Task 3 — RESTful Data Endpoints	Backend API	GET /api/v1/energy/live for the grid view and GET /api/v1/energy/live/{machine_id} for detail views; error handling (404 unknown machine, 503 database outage)
Task 4 — UI and Dashboard	Frontend	Machine Status Card (Active/Idle/Offline); 5-second auto-refresh; offline marking after 15 s of missing data; Chart.js visualization without memory leaks

Task 5 — QA and Integration	Testing	Mock IoT data generator; end-to-end latency verification under 5 s; API load testing with Postman/JMeter against simultaneous dashboard instances

5.2 SPRINT EXECUTION TIMELINE

Sprint 1 runs for two calendar weeks (ten working days), following the Scrum Guide cadence. The sprint backlog is refined into daily increments at the daily scrum, and progress is tracked visually through a burndown chart so that any deviation from the ideal trend is visible within a single day. Figure 6 shows the projected burndown for the 13 story points committed in Sprint 1; Figure 7 summarizes the story-point distribution across the four epics and the composition of the Sprint 1 scope.

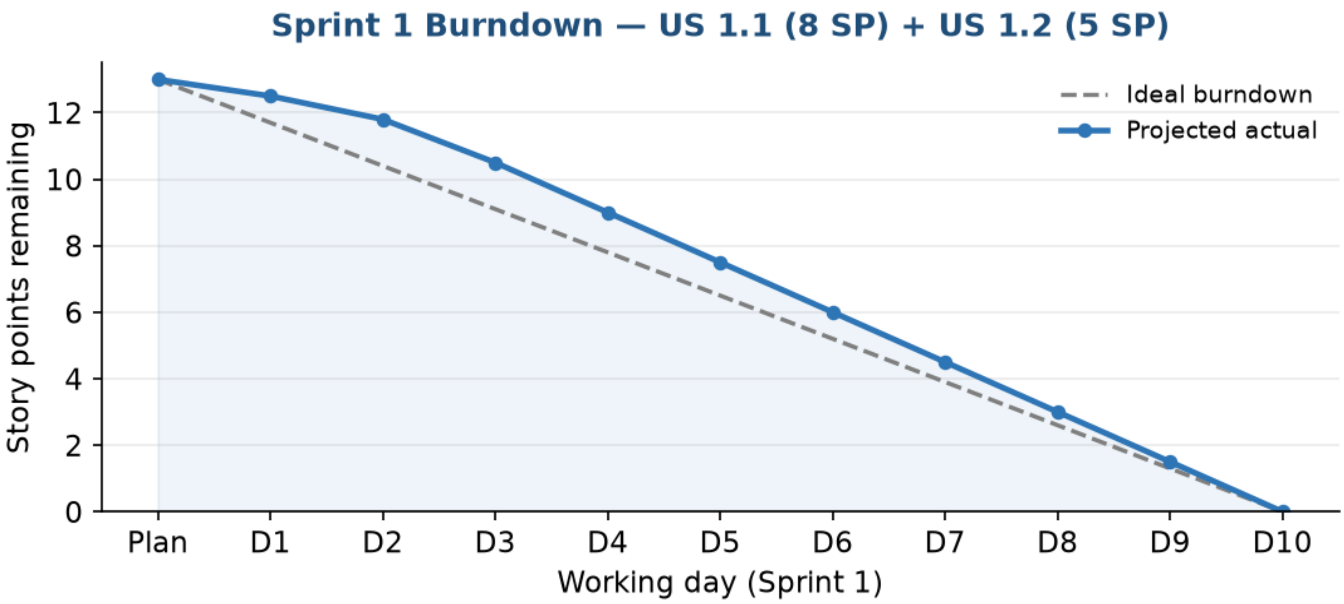
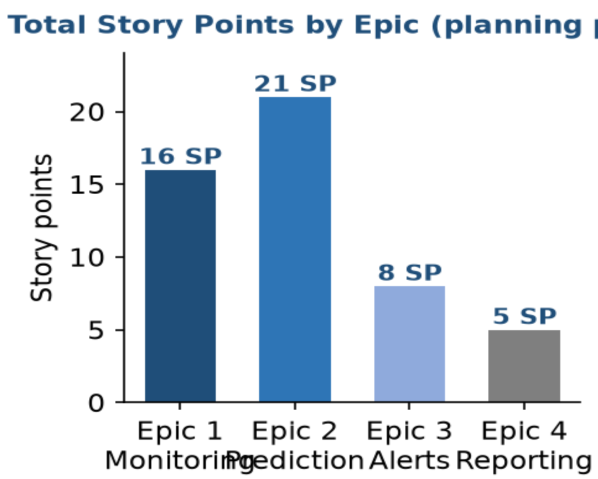


Figure 6 — Projected sprint burndown for Sprint 1 (13 story points): a slightly conservative actual trend that converges on the ideal burndown by the middle of the sprint, with two buffer days held back for integration and review preparation.



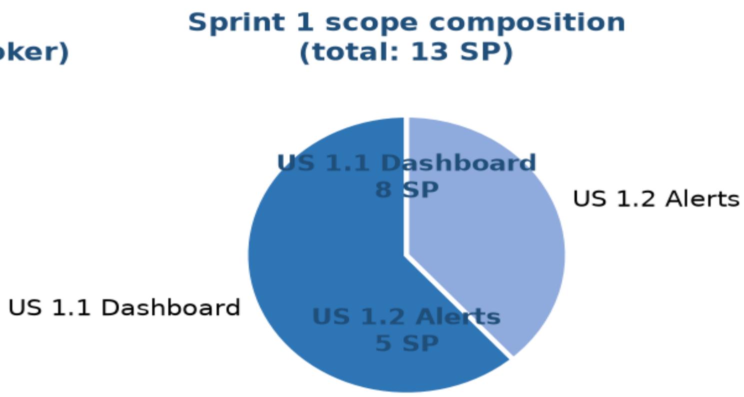


Figure 7 — Story-point distribution: the planning-poker totals across the four epics (left) and the composition of the Sprint 1 scope (right), where US 1.1 contributes 8 points and US 1.2 contributes 5.

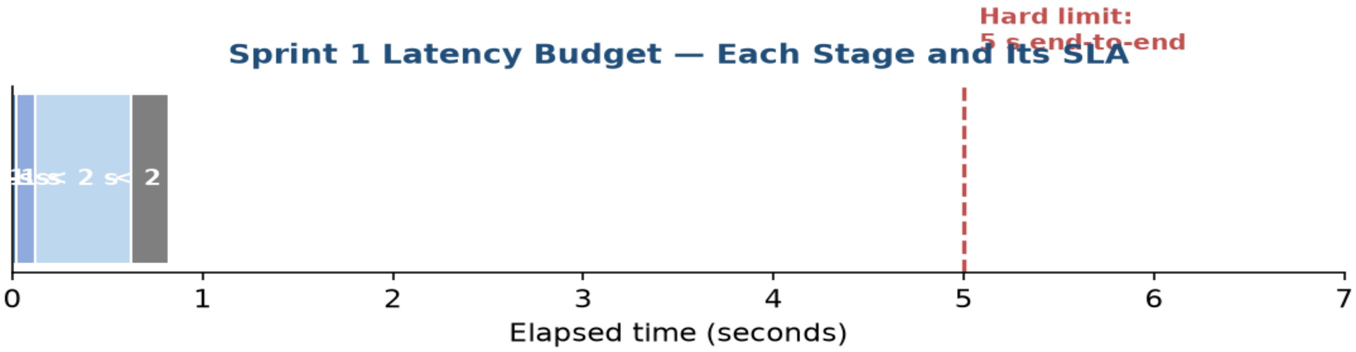


Figure 8 — Latency budget for US 1.1: the cumulative five-second end-to-end SLA is allocated across the sensor read, MQTT publish, database write, API response, and UI render stages.

5.3 STORY POINT ESTIMATION

Estimations used the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21) during a Planning Poker session.

User Story	Story Points	Justification
US 1.1 Dashboard	8	High complexity: full initial stack — database setup, MQTT pipeline, API, and UI
US 1.2 Automated Alerts	5	Medium complexity: reuses the US 1.1 pipeline but adds threshold logic and an SMS/push gateway
US 2.1 Forecasting	13	Very high complexity: data cleaning, feature engineering, LSTM training, plus forecast UI
US 3.1 Reporting	3	Low complexity: queries existing data into a downloadable PDF/CSV template

Sprint 1 carries US 1.1 (8 points) and US 1.2 (5 points), a total of 13 points, which is deliberately conservative for a first sprint in which the team is still calibrating its velocity. The remaining stories in the table above were estimated during the same Planning Poker session to give the Product Owner a full-picture roadmap even though they are not committed to Sprint 1.

5.4 SPRINT RISK REGISTER

In line with the Scrum Guide's emphasis on inspection and adaptation, the team maintains a short risk register that is reviewed at every daily scrum and formally revisited during the sprint retrospective. Each risk is rated by probability and impact, assigned a mitigation owner, and tracked until it is closed. The register below captures the four principal risks identified during Sprint 1 planning.

Risk	Probability	Impact	Mitigation	Owner
Sensor hardware delivery delayed by the vendor	Medium	High	Develop against a mock IoT data generator (Task 5) so dashboard work proceeds in parallel	Backend
MQTT message volume exceeds broker capacity	Low	High	Load-test Mosquitto with JMeter during Task 5; enable broker clustering before production	Data Eng.
Chart.js memory leak at 5-second refresh rate	Medium	Medium	Enforce chart destruction in the US 1.1 acceptance tests; browser profiling in QA	Frontend
SMS gateway API quota exhaustion during alert storms	Medium	Medium	Enforce the 15-minute cooldown (US 1.2); batch SMS via Twilio notify queue	Backend API

6. SPRINT GOAL AND EXPECTED DELIVERABLES

6.1 SPRINT GOAL

Sprint 1 Goal: Establish a secure, end-to-end data pipeline from factory floor sensors to the cloud, and deliver a functional real-time energy monitoring dashboard for at least one primary manufacturing machine.

The strategic justification is that this goal tackles the highest-risk technical dependency first — the hardware-to-cloud pipeline — while simultaneously delivering immediately visible business value to the Plant Manager. Validating that data can be ingested securely and visualized with minimal latency establishes a solid architectural foundation for the AI and forecasting features planned for future sprints.

6.2 EXPECTED DELIVERABLES

The expected Sprint 1 deliverables are, first, a **deployed time-series database and ingestion pipeline**: a configured InfluxDB or TimescaleDB instance in staging, a secure MQTT broker actively authenticating simulated sensor payloads, and verifiable data tables storing machine ID, timestamps, and energy metrics. Second, a **live monitoring web dashboard (MVP)**: a browser-accessible frontend featuring dynamic line charts rendered through Chart.js or Grafana panels, an auto-refresh mechanism that pulls new data points every five seconds without browser reload, and a machine status indicator that turns red when the data connection drops. Third, **API documentation**: fully tested RESTful endpoints documented in Swagger/OpenAPI, including request parameters, JWT authorization headers, and standard JSON response schemas for both success and error cases such as 404 for a missing machine. Fourth, an **interactive sprint**

review demonstration with the Plant Manager and Product Owner, featuring a live simulation of a machine generating energy data that appears on the dashboard within the five-second latency tolerance, followed by stakeholder feedback collection to inform Sprint 2 grooming.

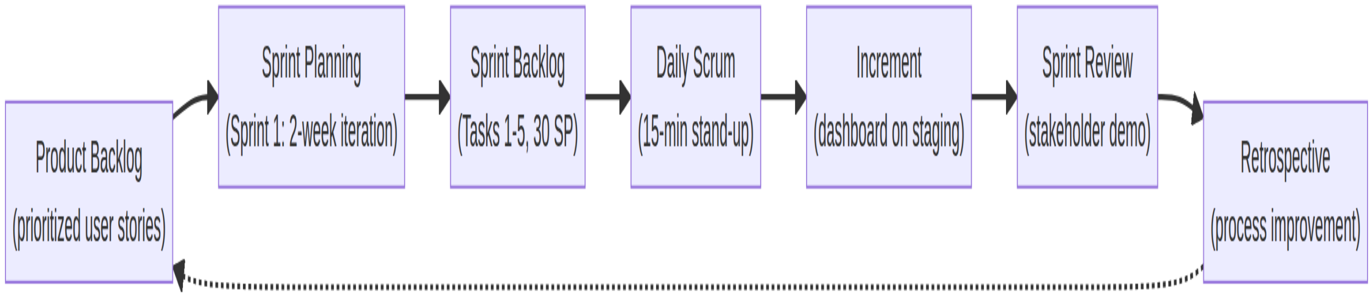


Figure 5 — The Scrum sprint cycle as executed for Sprint 1: from the prioritized product backlog through planning, the daily scrum, the increment, the sprint review, and the retrospective.

6.3 DEFINITION OF DONE

No item is marked complete for Sprint 1 unless all of the following are satisfied: the code is version-controlled, peer-reviewed, and merged into the main staging branch — in practice, feature branches reviewed through pull requests, consistent with a Gitflow strategy; unit and integration tests for the API and data pipeline are passing; the acceptance criteria for US 1.1 are verified by the QA lead; and the system is deployed and accessible in the staging environment, which itself runs on Docker containers for reproducibility between staging and production.

Beyond the Definition of Done, the team tracks sprint health through three lightweight metrics: **velocity** (story points delivered per sprint, expected to settle around 12–15 for this team), **burndown deviation** (difference between actual and ideal remaining points at each daily scrum), and **defect escape rate** (defects found after the sprint review per hundred story points). These metrics feed directly into the retrospective and inform how the committed 13 points of Sprint 2 are shaped, with US 1.4 (System Health) and US 1.3 (Granular Filtering) identified as the strongest Sprint 2 candidates because they deepen the platform the team will already have proven end-to-end.

6.4 LOOKING AHEAD TO SPRINT 2

Assuming the 13-point Sprint 1 velocity holds, Sprint 2 is provisionally scoped to approximately 15 story points and is expected to absorb US 1.4 (System Health), US 1.3 (Granular Filtering), and the entry point of Epic 3, namely US 3.1 (Instant Notifications), once the baseline energy-signature model established by US 1.2 has accumulated two weeks of production data. The Product Owner will confirm this scope at the Sprint 2 planning session using the feedback captured during the Sprint 1 review, ensuring the backlog remains aligned with both stakeholder priorities and demonstrated team capacity.

Sprint 2 Candidate	Epic	Story Points	Readiness Condition
US 1.4 System Health	1. Monitoring	5	Sensor registry from US 1.1 deployed in production
US 1.3 Granular Filtering	1. Monitoring	3	Department/shift tags populated in the time-series DB
US 3.1 Instant Notifications	3. Anomaly	8	Two weeks of baseline energy-signature data from US 1.2
US 3.2 Threshold Configuration	3. Anomaly	2	US 3.1 alert engine live; calibration feedback gathered
US 2.1 Peak Demand Forecast	2. Prediction	13	Thirty-plus days of historical consumption data

The conditional readiness column protects the team from over-committing: each Sprint 2 candidate enters the sprint backlog only when its data dependency is verifiably satisfied, preserving the team's ability to deliver its full commitment without mid-sprint scope collapse.

7. CONCLUSION

This report has established the complete Agile foundation for the AI-Driven Smart Industrial Energy Consumption Optimization Platform: a vision that bridges technical execution with business value, six stakeholder groups whose motivations shaped a thirteen-story backlog across four epics, rigorous BDD acceptance criteria for the highest-value stories, a MoSCoW-driven business-value prioritization, a concrete Sprint 1 backlog of five tasks estimated at 13 story points, a quantified risk register, and a well-justified sprint goal with an explicit Definition of Done and sprint health metrics.

By completing the data pipeline and real-time dashboard in Sprint 1, the team de-risks the project’s most critical technical dependency early and produces a demonstrable increment that earns stakeholder confidence for the predictive AI capabilities planned for subsequent sprints. The structured combination of epic-level decomposition, story-point estimation, and continuous inspection through burndown tracking and retrospectives ensures that each successive sprint compounds on verified learning rather than assumption, progressively transforming raw factory-floor telemetry into actionable, cost-saving, and compliance-grade intelligence.

REFERENCES

Schwaber, K., & Sutherland, J. (2020). The Scrum Guide. Scrum.org. https://scrumguides.org/

Cohn, M. (2005). Agile Estimating and Planning. Prentice Hall.

Kniberg, H. (2015). Scrum and XP from the Trenches. C4Media.

Beck, K. (1999). Extreme Programming Explained: Embrace Change. Addison-Wesley.